# Mathematical analysis 3
# Chapter 2 : Integral transformations
# Part 2: Laplace transform

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# Course outline

1. Definition and properties of Laplace
2. Inverse Laplace Transform
3. Applications

# Definition of Laplace transform

## Definition

*Let $t \mapsto f(t)$ be a function from $\mathbb{R}^+$ to $\mathbb{R}$. The Laplace transform of $f$, denoted by $L(f)$ or $\mathscr{L}(f)$, is defined by:*

$$L(f)(x) = \int_0^{+\infty} f(t)e^{-xt}\,dt.$$

## Remark

1. *We also use the notation $F$ to denote $L(f)$. $F$ is called the image of $f$ and $f$ is called the original of $F$.*
2. *The Laplace transform is defined in the same way for complex $x$.*

## Definition of Laplace transform

**Example.**

*If $U(t) = 1$ for $t \geq 0$ and is zero for $t < 0$ (called the unit step function or Heaviside function), then the Laplace transform of $U$ is defined if $x > 0$ and we have:*

$$L(U)(x) = \int_0^{+\infty} e^{-xt}\,dt = \frac{1}{x}.$$

# Existence

The function $F$ only exists if the integral defining it is convergent.

## Definition

*A function $f$ is said to be of exponential order at infinity if for sufficiently large $t$ we have:*

$$|f(t)| \leq Me^{\rho t},$$

*where $\rho \in \mathbb{R}$ and $M > 0$ are constants.*

## Practical remark

1. $f$ is of exponential order at infinity if and only if there exists $k \in \mathbb{R}$ such that $\lim\limits_{t \to +\infty} e^{-kt} f(t) = 0$.
2. If the constants $M$ and $\rho$ exist, they are not unique.

# Existence

**Example:**
Bounded functions (such as sine and cosine) are of exponential order at infinity. Indeed, if $f$ is bounded by a constant $C$, then

$$|f(t)e^{-t}| \leq Ce^{-t} \to 0, \text{ as } t \to +\infty.$$

# Existence

## Theorem

*If* $f$ *satisfies the following two conditions:*

- $f$ *is piecewise continuous on every interval* $[0,L] \subset [0,+\infty[$,
- $f$ *is of exponential order* $\rho$ *at infinity,*

*then the Laplace transform of* $f$ *exists for all* $x > \rho$ *and* $\lim\limits_{x\to+\infty} L(f)(x) = 0$.

# Abscissa of convergence

### Theorem

*Suppose there exists* $x \in \mathbb{R}$ *such that* $\int_0^{+\infty} f(t)e^{-xt}\,dt$ *converges.*

*Then, for any* $x_0 > x$, $\int_0^{+\infty} f(t)e^{-x_0 t}\,dt$ *also converges.*

### Theorem (Abscissa of convergence)

*Let* $f$ *be a function from* $\mathbb{R}^+$ *to* $\mathbb{R}$ *that is continuous. Then, there exists a unique* $\rho \in \mathbb{R}$ *(called Abscissa of convergence) such that*

$$x > \rho \Rightarrow \int_0^{+\infty} f(t)e^{-xt}\,dt \text{ converges } (L(f) \text{ exists}).$$

$$x < \rho \Rightarrow \int_0^{+\infty} f(t)e^{-xt}\,dt \text{ diverges } (L(f) \text{ does not exist}).$$

# Examples

For $f(t) = 0$ for all $t < 0$

| L'origine $f$ en $t$ | L'image $F = \mathscr{L}(f)$ en $x$ | Conditions d'existence |
|---|---|---|
| $1$ | $\frac{1}{x}$ | $x > 0$ |
| $t^n$ | $\frac{n!}{x^{n+1}}$ | $x > 0$ |
| $\sqrt{t}$ | $\frac{1}{2}\sqrt{\frac{\pi}{x^3}}$ | $x > 0$ |
| $\frac{1}{\sqrt{t}}$ | $\sqrt{\frac{\pi}{x}}$ | $x > 0$ |
| $e^{at}$ | $\frac{1}{x-a}$ | $x > a$ |
| $te^{at}$ | $\frac{1}{(x-a)^2}$ | $x > a$ |
| $\cos(at)$ | $\frac{x}{x^2+a^2}$ | $x > 0$ |
| $\sin(at)$ | $\frac{a}{x^2+a^2}$ | $x > 0$ |
| $\mathrm{ch}(at)$ | $\frac{x}{x^2-a^2}$ | $x > \|a\|$ |
| $\mathrm{sh}(at)$ | $\frac{a}{x^2-a^2}$ | $x > \|a\|$ |

# Properties of Laplace Transform

## Proposition (Linearity)

*Let $f$ and $g$ be two causal functions admitting Laplace transforms $L(f)$ and $L(g)$ respectively, such that*

- $L(f)$ *exists for* $x > \rho_1$,
- $L(g)$ *exists for* $x > \rho_2$.

*Then, for any* $\alpha, \beta \in \mathbb{R}$, *we have*

$$L(\alpha f + \beta g)(x) = \alpha L(f)(x) + \beta L(g)(x), \forall x > \max(\rho_1, \rho_2)$$

**Example:** Calculate the Laplace transform of the function

$$f(t) = 3\cos(t) - 2t.$$

We have

$$L(f)(x) = 3L(\cos(t))(x) - 2L(t)(x) = \frac{3x}{x^2+1} - \frac{2}{x^2}, \quad \forall x > 0.$$

# Properties of Laplace Transform

## Proposition (Laplace Transform of Translation)

*For any* $\alpha > 0$, *we have* $L(f(t-\alpha))(x) = e^{-\alpha x} F(x)$, *for all* $x > \rho$.

**Example:** Since $L(\sin(t))(x) = \dfrac{1}{x^2+1}$ for all $x > 0$, then

$$L(\sin(t-3))(x) = \frac{e^{-3x}}{x^2+1}, \text{ for all } x > 0$$

.

## Proposition (Translated Laplace Transform)

*For any* $\alpha \in \mathbb{R}$, *we have* $L(e^{\alpha t} f(t))(x) = F(x-\alpha)$, *for all* $x > \rho + \alpha$.

**Example:** Since $L(\cos(t))(x) = \dfrac{x}{x^2+1}$ for all $x > 0$, then

$$L(e^{2t}\cos(t))(x) = \frac{x-2}{(x-2)^2+1}, \quad \forall x > 2$$

# Properties of Laplace Transform

## Proposition

*For any* $\alpha > 0$,

$$L(f(\alpha t))(x) = \frac{1}{\alpha} F\left(\frac{x}{\alpha}\right), \forall x > \alpha \rho$$

.

**Example:** Since $L(\sin(t))(x) = \dfrac{1}{x^2+1}$ for all $x > 0$, then

$$L(\sin(2t))(x) = \frac{2}{x^2+4}, \forall x > 0$$

. Since $L(\cos(t))(x) = \dfrac{x}{x^2+1}$ for all $x > 0$, then

$$L(\cos(3t))(x) = \frac{x}{x^2+9}, \forall x > 0$$

.

# Properties of Laplace Transform

## Proposition (Laplace Transform of the Derivative)

*Let $f$ be a function from $\mathbb{R}^+$ to $\mathbb{R}$ such that*

- *$f$ is piecewise continuous on each interval $[0,N] \subset \mathbb{R}^+$.*
- *$f$ is of exponential order at the infinity.*
- *$f$ differentiable on $\mathbb{R}^+$.*
- *$f'$ piecewise continuous on each interval $[0,N] \subset \mathbb{R}^+$*

*and if $L(f)(x) = F(x)$ for all $x > \rho$. Then*

$$L(f'(t))(x) = xF(x) - f(0^+), \forall x > \rho$$

.

# Properties of Laplace Transform

## Proposition (Generalization)

*Let $f$ be a function from $\mathbb{R}^+$ to $\mathbb{R}$.*

- *For all $k \in \{0, \ldots, n\}$, $f^{(k)}$ exists and is piecewise continuous on $\mathbb{R}^+$.*
- *For all $k \in \{0, \ldots, n-1\}$, there exist $M$ and $\rho \in \mathbb{R}$ such that $|f^{(k)}(t)| \leq Me^{\rho t}$ for $t$ large enough.*

*Then, for all $x > \rho$, we have*

$$L(f^{(n)}(t))(x) = x^n F(x) - x^{n-1} f(0^+) - \ldots - x f^{(n-2)}(0^+) - f^{(n-1)}(0^+).$$

# Properties of Laplace Transform

## Proposition (Derivative of the Laplace Transform)

*If* $f:\mathbb{R}^+ \to \mathbb{R}$ *such that* $L(f)(x) = F(x)$ *for* $x > \rho$, *then* $F$ *is* $C^\infty$ *on* $]\rho, +\infty[$ *and for all* $n \in \mathbb{N}$ *we have*

$$F^{(n)}(x) = (-1)^n L(t^n f(t))(x), \quad \forall x > \rho.$$

**Example.** Calculate $L(t^2 \sin(t))$.

We set : $f(t) = \sin(t)$, then we have $F(x) = L(f(t)) = \frac{1}{x^2+1}$ for $x > 0$. So

$L[t^2 \sin(t)] = (-1)^2 F''(x) = \left(\frac{1}{(x^2+1)}\right)''$ for $x > 0$.

$L[t^2 \sin(t)] = \dfrac{6x^2 - 2}{(x^2+1)^3}$ for $x > 0$.

# Properties of Laplace Transform

## Proposition (Integration of the Laplace Transform)

*If* $f:\mathbb{R}^+\to\mathbb{R}$ *such that* $L(f)(x)=F(x)$ *for* $x>\rho$ *and* $\lim_{t\to 0^+}\frac{f(t)}{t}$ *is finite, then*

$$\int_x^{\infty} F(s)\,ds = L\left(\frac{f(t)}{t}\right)(x), \forall x>\rho.$$

## Proposition (Transform of the Primitive)

*Let* $f$ *be a function from* $\mathbb{R}^+$ *to* $\mathbb{R}$ *such that* $L(f)=F$ *for* $x>\rho$. *Then,*

$$L\left(\int_0^t f(s)\,ds\right)(x)=\frac{F(x)}{x}, \forall x>\max(0,\rho).$$

# Properties of Laplace Transform

## Proposition

*If $f$ is a $T$-periodic function, defined, continuous, and bounded on $\mathbb{R}^+$, then*

$$L(f)(x) = \frac{\int_0^T f(t)e^{-xt}}{1-e^{-xT}}\,dt, \forall x > \max(0,\rho).$$

# Initial and final value

## Proposition

*Suppose* $f$ *and* $f'$ *are piecewise continuous on the interval* $[0, L]$ *and of exponential order.*

*1) If* $f$ *has a Laplace transform* $L(f) = F$ *for* $x > \rho$, *then*

$$\lim_{x \to +\infty} xL(f)(x) = \lim_{t \to 0} f(t).$$

*2) If* $f$ *has a Laplace transform* $L(f) = F$ *for* $x > \rho$ *with* $\rho < 0$, *and* $\lim_{t \to +\infty} f(t)$ *exists, then*

$$\lim_{x \to 0} xL(f)(x) = \lim_{t \to +\infty} f(t).$$

# Laplace transform of the convolution product

## Definition

*The convolution product of two functions* $f$ *and* $g$ *is defined by:*

$$(f*g)(t)=\int_{-\infty}^{\infty} f(u)g(t-u)\,du$$

## Proposition (Laplace Transform of the Convolution Product)

*Let* $f$ *and* $g$ *be two functions from* $\mathbb{R}^+$ *to* $\mathbb{R}$ *such that* $L(f)=F$ *and* $L(g)=G$ *respectively for* $x>\rho_1$ *and* $x>\rho_2$ *respectively. Then,*

$$L(f*g)(x)=F(x)G(x),\forall x>\max(\rho_1,\rho_2)$$

# Course outline

# Inverse Laplace transform

## Definition

*Let $f$ be a causal function (i.e., $f(t) = 0$ for $t < 0$) admitting a Laplace transform $L(f) = F$ for $x > \rho$. Then, the original function $f$ is called the inverse Laplace transform of $F$, denoted as $L^{-1}(F) = f$.*

**Example:** Let $F(x) = \frac{1}{x-3}$. Calculate $L^{-1}(F)$ if it exists.
Since $L^{-1}(e^{3t})(x) = \frac{1}{x-3}$, we have $L^{-1}(F)(t) = e^{3t}$ for $t \geq 0$.
Thus, $L^{-1}(F)(t) = e^{3t}U(t)$, where $U$ is the unit step function defined by $U(t) = 1$ for $t \geq 0$ and zero for $t < 0$.
**Remark:** There are functions $F$ that do not admit an inverse Laplace transform.

# Linearity of the inverse of Laplace transform

## Proposition

*Let $F$ and $G$ be two causal functions such that $L^{-1}(F) = f$ and $L^{-1}(G) = g$, where $f$ and $g$ are continuous on $\mathbb{R}^+$ and of exponential order. Then, for all $\alpha, \beta \in \mathbb{R}$, $L^{-1}(\alpha F + \beta G) = \alpha L^{-1}(F) + \beta L^{-1}(G)$.*

**Example:** Calculate $L^{-1}\left(\frac{1}{x^3+x}\right)$ if it exists.

Since

$$\frac{1}{x^3+x} = \frac{1}{x(x^2+1)} = \frac{1}{x} - \frac{x}{x^2+1},$$

we have $L^{-1}\left(\frac{1}{x^3+x}\right)(t) = L^{-1}\left(\frac{1}{x}\right)(t) - L^{-1}\left(\frac{x}{x^2+1}\right)(t) = (1-\cos(t))U(t)$.

The following result allows finding the inverse Laplace transform of a rational fraction $\frac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are polynomials with $\deg(P) < \deg(Q)$.

**Proposition (Heaviside's Formula):** Let the fraction be irreducible, with $Q$ of degree $n$ and having only simple and distinct roots $r_1, \ldots, r_n$. Then,

$$L^{-1}\left(\frac{P(x)}{Q(x)}\right)(t) = \frac{P(r_1)}{Q'(r_1)}e^{r_1 t} + \ldots + \frac{P(r_n)}{Q'(r_n)}e^{r_n t}, \quad \forall t \geq 0.$$

**Example:** Calculate $L^{-1}\left(\frac{1}{x^2-3x+2}\right)$. Using Heaviside's formula,

$$L^{-1}\left(\frac{1}{x^2-3x+2}\right)(t) = L^{-1}\left(\frac{1}{(x-1)(x-2)}\right)(t) = (-e^t + e^{2t})U(t).$$

# Other properties of Laplace transform

## Proposition

*If* $L^{-1}(F) = f$, $L^{-1}(G) = g$, *and* $f$, $g$ *are continuous on* $\mathbb{R}^+$, *and of exponential order, then:*

1. $L^{-1}\left(e^{-\alpha x}F(x)\right) = f(t-\alpha)$ *with* $\alpha > 0$,
2. $L^{-1}(F(x-\alpha)) = e^{\alpha t}f(t)$ *with* $\alpha \in \mathbb{R}$,
3. $L^{-1}\left(F\left(\frac{x}{\alpha}\right)\right) = \alpha f(\alpha t)$ *with* $\alpha > 0$,
4. *If* $f(0) = 0$, *then* $L^{-1}(xF(x)) = f'(t)$,
5. $L^{-1}\left(F^{(n)}(x)\right) = (-1)^n t^n f(t)$,
6. $L^{-1}\left(\int_x^{\infty} xF(s)\,ds\right) = \frac{f(t)}{t}$,
7. $L^{-1}\left(\frac{F(x)}{x}\right) = \int_0^t f(s)\,ds$,
8. $L^{-1}(FG) = L^{-1}(F) * L^{-1}(G) = f * g$.

# Inverse of Laplace transform

**Theorem (Lerch's Uniqueness)**

*If $f$ and $g$ are two functions continuous on $\mathbb{R}_+$ such that*

$$L(f(t)) = L(g(t)),$$

*then*

$$f(t) = g(t) \ \ \forall t \in \mathbb{R}_+.$$

# Course outline

# Resolution of ODEs

**Example:** Consider finding a function $f$ satisfying $f'' - 3f' + 2f = 1$, $t \geq 0$ with $f(0) = f'(0) = 0$.

**Solution:** Applying the Laplace transform to both sides of the ODE, we get (using linearity and differentiation properties of the Laplace transform):

$$L(f'' - 3f' + 2f)(x) = L(1)(x)$$

$$\Rightarrow x^2 L(f) - xf(0) - f'(0) - 3(xL(f) - f(0)) + 2L(f) = L(1)$$

$$\Rightarrow x^2 L(f) - 3xL(f) + 2L(f) = L(1)$$

$$\Rightarrow (x^2 - 3x + 2)L(f) = \frac{1}{x}$$

$$\Rightarrow L(f)(x) = \frac{1}{x(x^2 - 3x + 2)} = \frac{1}{2x} - \frac{1}{(x-1)} + \frac{1}{2(x-3)}$$

$$\Rightarrow f(t) = \frac{1}{2} L^{-1}\left(\frac{1}{x}\right) - L^{-1}\left(\frac{1}{x-1}\right) + \frac{1}{2} L^{-1}\left(\frac{1}{x-2}\right)$$

$$\Rightarrow f(t) = \frac{1}{2} - e^t + \frac{1}{2} e^{2t}, \quad t \geq 0$$

**Example:** Find a function $f$ satisfying $f'' + f = t$, $t \geq 0$ with $f(0) = f'(0) = 0$.

**Solution:** Applying the Laplace transform to both sides of the ODE, we get (using linearity and differentiation properties of the Laplace transform):

$$
\begin{aligned}
L(f'' + f)(x) = L(t)(x) \Rightarrow & x^2 L(f) + L(f) = \frac{1}{x^2} \\
& \Rightarrow x^2 L(f) + L(f) = \frac{1}{x^2} \\
& \Rightarrow (x^2 + 1) L(f) = \frac{1}{x^2} \\
& \Rightarrow L(f)(x) = \frac{1}{x^2(x^2 + 1)} \\
& \Rightarrow L(f)(x) = \frac{1}{x^2} - \frac{1}{x^2 + 1} \\
& \Rightarrow f(t) = t - \sin(t), \quad t \geq 0
\end{aligned}
$$

**Example** Using Laplace transforms, we are interested in solving the following system of ODEs:

$$\begin{cases} y'' + (z' - y') = -\frac{3}{4}y \text{ on } \mathbb{R}_+, \\ z'' - (z' - y') = -\frac{3}{4}z \text{ on } \mathbb{R}_+, \end{cases}$$

with

$$y(0) = z(0) = 0, \quad y'(0) = 1, z'(0) = -1.$$

**Solution:** Let $Y = L(y)$, $Z = L(z)$, and apply the Laplace transform to the given system:

$$\begin{cases} L(y'') + (L(z') - L(y'))) = -\frac{3}{4}L(y), \\ L(z'') - (L(z') - L(y'))) = -\frac{3}{4}L(z), \end{cases}$$

which simplifies to

$$\begin{cases} x^2Y - xy(0) - y'(0) + (xZ - z(0) - xY + y(0)) = -\frac{3}{4}Y, \\ x^2Z - xz(0) - z'(0) - (xZ - z(0) - xY + y(0)) = -\frac{3}{4}Z, \end{cases}$$

$$\begin{cases} x^2Y - 1 + (xZ - xY) = -\frac{3}{4}Y, \\ x^2Z + 1 - (xZ - xY) = -\frac{3}{4}Z. \end{cases} \Rightarrow \begin{cases} x^2Y - 1 + x(Z - Y) = -\frac{3}{4}Y, \ldots (1) \\ x^2Z + 1 - x(Z - Y) = -\frac{3}{4}Z, \ldots (2) \end{cases}$$

$$\Rightarrow \begin{cases} x^2(Y + Z) = -\frac{3}{4}(Y + Z), \ldots (1) + (2) \\ x^2(Y - Z) - 2 + 2x(Z - Y) = -\frac{3}{4}(Y - Z), \ldots (1) - (2) \end{cases}$$

$$\Rightarrow \begin{cases} (x^2 + \frac{3}{4})(Y + Z) = 0, \\ x^2Z + 1 - x(Z - Y) = -\frac{3}{4}Z, \end{cases} \Rightarrow \begin{cases} Y + Z = 0, \\ x^2Z + 1 - x(Z - Y) = -\frac{3}{4}Z, \end{cases}$$

$$\Rightarrow \begin{cases} Y = -Z, \\ x^2Z + 1 - x(Z - Y) = -\frac{3}{4}Z, \end{cases}$$

$$\Rightarrow \begin{cases} Z = -Y, \\ -x^2 Y + 1 + 2xY = \frac{3}{4} Y \end{cases} \quad \Rightarrow \begin{cases} Z = -Y, \\ -4x^2 Y + 8xY - 3Y = -4 \end{cases}$$

$$\Rightarrow \begin{cases} Z = -Y, \\ (4x^2 - 8x + 3) Y = 4 \end{cases} \quad \Rightarrow \begin{cases} Z = -Y, \\ Y = \frac{4}{4x^2 - 8x + 3}. \end{cases}$$

We have $4x^2 - 8x + 3 = 4(x - \frac{1}{2})(x - \frac{3}{2})$, so $Y = \frac{1}{(x-\frac{1}{2})(x-\frac{3}{2})}$. Thus,
$Y = -\frac{1}{x-\frac{1}{2}} + \frac{1}{x-\frac{3}{2}}$ and $y(t) = -L^{-1}\left(\frac{1}{x-\frac{1}{2}}\right) + L^{-1}\left(\frac{1}{x-\frac{3}{2}}\right)$.
This implies

$$\begin{cases} y(t) = -e^{\frac{t}{2}} + e^{\frac{3t}{2}}, \quad \forall t \geq 0 \\ z(t) = -y(t) = e^{\frac{t}{2}} - e^{\frac{3t}{2}}, \quad \forall t \geq 0. \end{cases}$$